Assignment

1. Write a program that will take a student's name (in a string) and five exam marks (in a float array) as input in main function. Then define two functions with the following tasks:
   a. Detect the last name of the student(last word of the string) and print it.
   b. Find the average exam marks and return the value to main.

   Sample Input:

   Enter student name :Ruhan Khan
   Enter five exam marks : 15.5 16.5 14 20 19

   Sample output:

   The Last name is : Khan
   Average marks is : 17.0

2. Declare a structure date with year, month and day (all integer) as its attributes. Take five dates as input.
   - Print the stored data as output.
   - Print the dates that fall on a leap year in the format dd/mm/yyyy.

   All outputs should be shown in the console.

3. Eleven is overwhelmed by the recent outbreak of Demogorgon unleashed by Vecna. Every Demogorgon is marked with a specific number. Write a Program that will detect all Demogorgon Numbers in between TWO Ranges given by Eleven.
   Demogorgon numbers are those numbers that have the following CONDITIONS:
   [CONDITION 1] Sum of ODD digits is less than the Sum of EVEN digits in that number.
   [CONDITION 2] It is a Prime number

   Write following functions to solve this problem:
   int DemogorgonNumber (int a): if DEMOGORGON returns 1 else returns -1
   int checkerFunc1 (int a): if CONDITION No. 1 is true returns 1 else returns -1
   int primeChecker (int a): if CONDITION No. 2 is true returns 1 else returns -1

Sample Input	Sample Output
130 430	**163 181 223 241 263 281 283 383 401 421**
1005 1250	**1061 1163 1181**

4. Ms. Isabella wants to find out her best 3 loyal customers. She has the following data of the customer:
   • Customer ID
   • Name
   • Birthday: dd/mm/yyyy
   • House: Gryffindor, Slytherin
   • Purchase Cost of Last 3 Months

For Example, 11034, Christopher Nolan, 11/06/1945, Gryffindor, [40, 56, 67]
Loyal Customer will be awarded to the person with the highest average purchase value from each House with the following additional criteria:

• If House is Gryffindor, he/she must contain **"est"** in his name.
• If House is Slytherin, he/she must contain **"rus"** in her name.

Write a Program to assist Ms. Isabella. Your Program must contain a structure that will be able to contain the data of a customer. If There is No Loyal Customer from a House Print Nil. Else Print the Customer Name, Birthday, and Average Purchase.

NOTE THAT - You must write a function named **int substringChecker (char*mainString, char *searchString)** and utilize that while searching a certain sub-string in another string**. [The Function must be called by Reference only]**

Input	Output	Explanation
Num of Customer: 3 Customer 1: Name: Harry Potter Birthday: 11/01/1985 House: Gryffindor Purchase: 22 45 65 Customer 2: Name: Albus Severus Potter Birthday: 11/04/1987 House: Slytherin Purchase: 22 45 65 Customer 3: Name: Hermione Grenzer Birthday: 11/04/1985 House: Gryffindor Purchase: 21 42 65 Customer 4: Name: Severus Snape Birthday: 23/04/1967 House: Slytherin Purchase: 34 41 95 Customer 5: Name: Draco Malfoy Birthday: 11/04/1985 House: Slytherin Purchase: 100 41 45	List of Loyal Customers: Gryffindor: Nil Slytherin: Severus Snape | 23/04/1967 | 56.67	Both the Gryffindor Customers Violate the constraint thathe/she must contain **"est"** in his name. Thus NIL. In Slytherin, the average purchase value is as follows: Draco Malfoy | 62 Severus Snape | 56.67 Albus Severus Potter | 44 Though Draco has the highest average, he doesn't have**"rus"** in his name. Thus, the Loyal customer from Slytherin is Severus Snape.